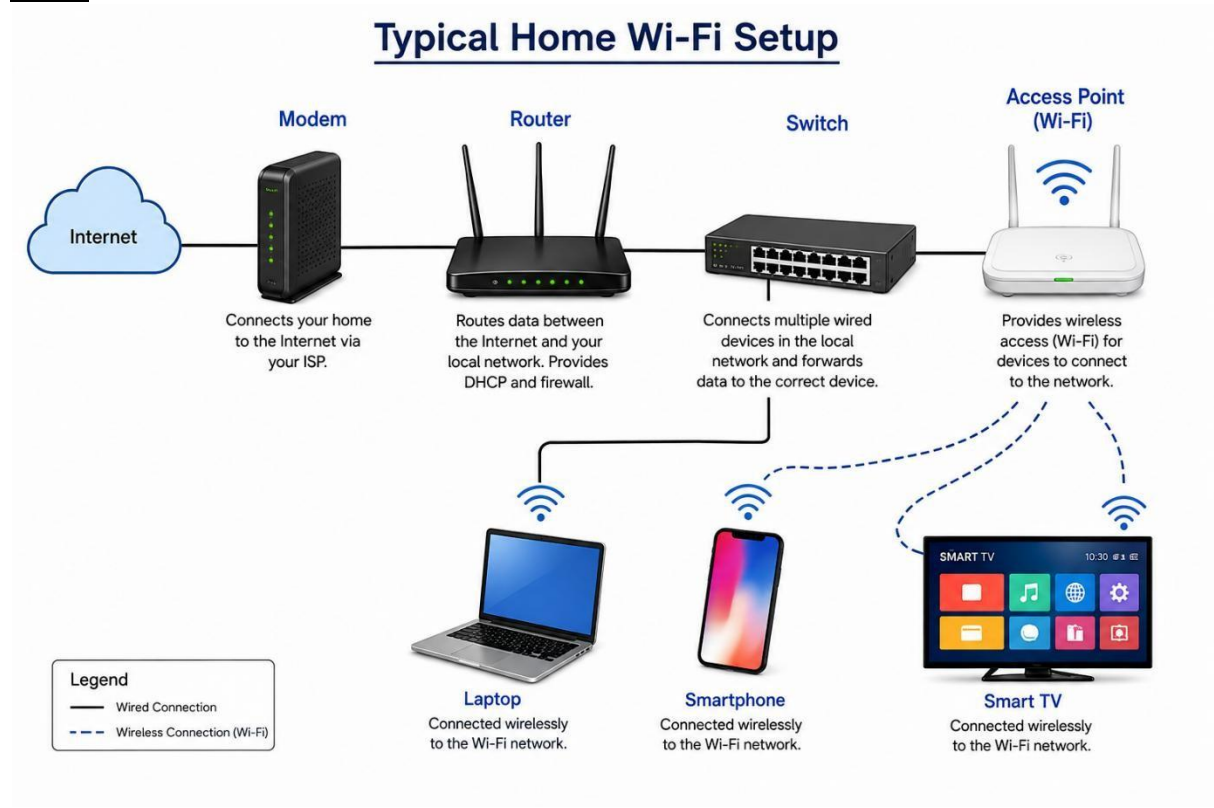


1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

ANS:-



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

ANS:-

Networking Device	Main Function	Real-Life Example
Modem	Connects your home network to the Internet through the ISP.	Used at home to receive Internet service from an ISP.
Router	Routes data between different networks and shares the Internet with multiple devices.	Used at home to connect phones, laptops, and TVs to the Internet.
Switch	Connects multiple devices within the same local network and forwards data to the correct device.	Used in offices to connect computers, printers, and servers.

Access Point (AP)	Provides wireless Wi-Fi access to devices.	Used in offices, colleges, and homes to provide Wi-Fi coverage.
Network Cable	Physically connects network devices and carries data between them.	Ethernet cable used to connect a computer to a router or switch.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

ANS:-

Cable Type

Common Use

Commonly used to connect computers, routers, switches, and other devices in a LAN.

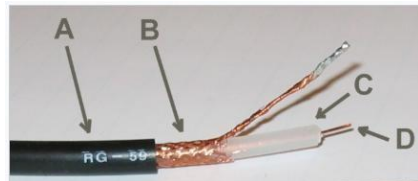
Ethernet Cable (Twisted Pair)



[Ethernet physical layer - Wikipedia](#)

Commonly used for cable TV and some broadband Internet connections.

Coaxial Cable



[Coaxial cable - Wikipedia](#)

Used for high-speed Internet, telecommunications, and long-distance data transmission.

Fiber Optic Cable



[Optical fiber - Wikipedia](#)

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

ANS:-

Feature	Router	Switch
1. Main Use	Connects different networks, such as a home network to the Internet.	Connects multiple devices within the same network.
Feature	Router	Switch
2. Typical Location	Usually placed at the edge of a network, near the modem/ISP connection.	Usually used inside a LAN, such as an office or computer lab.
3. How It Handles Data	Uses IP addresses to route packets between networks.	Uses MAC addresses to forward frames to the correct device.
4. Internet Connection	Can provide Internet access to multiple devices.	Normally does not directly provide Internet access; it connects devices to the LAN.
5. Example	Home Wi-Fi router connecting phones and laptops to the Internet.	Ethernet switch connecting computers and printers in an office.